PAPER

When simple wins: leakage-free benchmarking of classical Machine learning vs. graph neural networks in low-data bioactivity prediction

Abstract

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1 Introduction

1.1 Predictive Modeling in Low-Data Bioactivity Regimes

It is a well-established dogma in quantitative structure-activity relationships (QSAR) that parsimonious models generalize best [1]. In data-limited scenarios, simpler interpretable approaches like multivariate linear regression have traditionally prevailed, often demonstrating prediction reliability equivalent to complex artificial neural networks [2, 3]. These linear models maintain higher bias and lower variance, avoiding the extreme sensitivity to random fluctuations seen in complex algorithms, especially when supplemented by rigorous quantum chemical calculations to derive robust descriptors [3].

Modeling small datasets is inherently susceptible to overfitting. This occurs either by employing an overly flexible architecture or by including irrelevant predictors that add random variation to predictions [4]. This bottleneck is severely exacerbated when the high dimensionality of molecular descriptors vastly exceeds the small sample size [5]. Furthermore, small datasets lack the statistical tolerance of large numbers, making model performance acutely sensitive to data bias and minor descriptor inaccuracies, which can severely hamper the effectiveness and interpretability of virtual screening [6, 2]. Conversely, when descriptors are meticulously curated, machine learning can effectively guide rational design, successfully discriminating potent bioactives—such as Amyloid-β aggregation inhibitors—from decoys [7].

Applying complex, non-linear architectures to these regimes introduces profound validation hurdles. Building neural networks from scratch typically risks catastrophic overfitting, as complex models adapt to noise and generate deceptively overoptimistic internal metrics [8, 9]. This danger extends directly into model selection; tuning hyperparameters is exceptionally prone to overfitting when sample sizes are small and the parameter space is large [10]. Consequently, evaluating machine learning architectures in isolation is meaningless without rigorously assessing the combined model selection procedure, making double (or nested) cross-validation mandatory to ensure true independence between model building and selection [8, 10].

Traditional evaluation metrics and standard k-fold cross-validation often misleadingly overestimate explorative prediction capabilities [11, 12]. For instance, algorithms like Random Forest fundamentally lack extrapolation capability, as their ensemble averaging cannot predict values outside the range of the training data bounds [12]. Similarly, attempting to navigate unstructured, discrete chemical spaces with generative deep learning faces severe limitations with sparse data, where even a single-character alteration in a fragile SMILES string can irreparably break a legal molecular structure [13].

To safely harness the potential of overparameterized models [1] without falling into the overfitting trap, rigorous validation protocols are mandatory. Techniques such as y-randomization (response shuffling) become essential to verify that the model captures true chemical relationships rather than random noise [14].

Finally, methodologies like transfer learning bypass data scarcity by adapting base networks previously optimized on large source domains [9]. By utilizing reduced learning rates during fine-tuning [15], and preferring efficient architectures like convolutional neural networks over heavier Transformer models [16], researchers can robustly navigate class imbalance and model selection bias in low-resource settings.

1.2 Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies heavily on domain-engineered descriptors and molecular fingerprints [17]. Methods like Morgan (ECFP4) encode structural traits, while topological, 3D, and physicochemical descriptors—such as radial distribution functions, Burden eigenvalues, and ab initio quantum chemical calculations—capture specific molecular interactions deterministically without requiring massive data [7, 18, 3]. This approach embeds theoretical chemistry directly into the model, ensuring high interpretability and clarity [3]. However, accumulating these handcrafted features creates highly dimensional, redundant matrices [5, 2]. In small datasets, this redundancy increases computational costs and severely worsens the feature selection bottleneck [5, 2].

Alternatively, end-to-end representation learning bypasses manual feature selection by autonomously mapping discrete molecular topologies into continuous latent spaces [19, 6]. Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—extract task-specific traits directly from chemical structures to model complex, nonlinear interactions [6, 17, 13]. Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES [16]. However, learning representations in discrete chemical spaces is intrinsically fragile; altering even a single character in a SMILES string can irreparably invalidate the entire molecular structure [13].

Despite these advantages, does deep learning complexity truly benefit low-data regimes? These algorithms are inherently data-hungry, making representation learning highly ineffective when ground-truth labels and dataset volumes are severely restricted [20]. Training complex networks from scratch on small datasets risks catastrophic overfitting, compromised generalization, and suboptimal training [2, 21, 9]. Consequently, simpler classical ensembles and highly structured linear models (e.g., Multiple Linear Regression) frequently achieve equivalent or superior predictive reliability compared to standalone artificial neural networks (ANNs) in data-scarce scenarios [18, 17, 3]. Because small datasets cannot naturally sustain deep architectures, researchers often limit studies to classical algorithms [11].

Nevertheless, traditional skepticism toward nonlinear models in limited-data scenarios is changing [22]. When properly tuned, regularized, and hybridized with structured linear priors or heuristics to stabilize training, nonlinear algorithms can achieve competitive results comparable to robust classical methods [22, 21]. Furthermore, transfer learning serves as a vital catalyst. By fine-tuning pretrained networks like ChemBERTa, researchers can achieve accurate representation learning predictions even with only dozens of target data points [23, 9].

1.3 Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage is the unintended introduction of test data information into the training process, a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings [24, 25]. A common trap involves overall decomposition or applying mathematical scaling across the entire dataset prior to partitioning, which illicitly leaks future information into the training phase [26]. To ensure true statistical independence, researchers must design leak-free pipelines that strictly encapsulate all preprocessing and scaling steps exclusively within the training folds after the test data has been safely separated [27]. Furthermore, rigorous partitioning protocols, such as a strict three-way split, must be enforced to guarantee there is absolutely no overlap among training, validation, and test sets across subjects or discrete samples [21].

Beyond preprocessing, data leakage frequently corrupts the model selection phase. Tuning hyperparameters globally across the entire dataset grants the algorithm illicit knowledge of the test distribution, yielding spuriously high performance [28]. Even in traditional internal validation, an overlap between the data used for hyperparameter tuning and the test set creates an optimistic bias that renders the evaluation uninformative regarding real-world generalizability [29]. The validation dataset fundamentally loses its independence if its collective predictions continuously influence the search for a good model, triggering severe model selection bias [8]. This lack of independence causes models—especially overly complex deep learning architectures deployed on small pretraining datasets [20]—to overfit by adapting to noise, producing deceptively overoptimistic internal metrics that severely underestimate true generalization errors [8].

To mitigate these structural biases, model selection must be treated as an integral, encapsulated component of the fitting process. Double or nested cross-validation is mandatory to generate strictly independent test objects in an outer loop that remain entirely isolated from model building and hyperparameter optimization [8]. To further prevent fortuitous or biased splitting, researchers should employ deterministic methods like the Kennard-Stone algorithm to construct highly representative, unbiased external test sets [3]. When low-data regimes necessitate data augmentation, methodological purity is paramount: classifier performance must be measured exclusively on real test data, keeping synthetic samples strictly isolated, while employing probabilistic Bayesian optimization rather than exhaustive grid searches to safely navigate the hyperparameter space [30]. Preserving and transparently documenting this absolute methodological rigor is critical; insufficient reporting of data partitioning often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage [31].

1.4 Objectives and Principal Contributions

This study bridges the gap between stringent statistical validation and advanced representation learning to accelerate targeted drug discovery in low-data regimes.

- **Methodological Reproducibility and Leak-Free Validation:** Implementation of a fully transparent, 100% reproducible Python pipeline built upon a rigorous 15-Fold Nested Cross-Validation protocol. This ensures absolute statistical independence between hyperparameter optimization and model evaluation, effectively eliminating model selection bias and data leakage.

- **Architectural Benchmarking in Data-Limited Regimes:** A comprehensive empirical comparison evaluating the predictive reliability of traditional domain-engineered classical ensembles against state-of-the-art Graph Neural Networks (specifically, the ChemProp Directed Message Passing Neural Network, D-MPNN) using a standardized reference dataset.

- **Translational Drug Repurposing Pipeline:** The seamless integration of machine learning predictions with structure-based Molecular Docking. This hybrid pipeline systematically screens, identifies, and prioritizes FDA-approved compounds, providing a validated computational pathway for rapid therapeutic repurposing.

2 Materials and Methods

2.1 Bioactivity Benchmark Dataset and Curation

The dataset used in this study was derived from a previously published benchmark on *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors [32]. The provenance of these data relies on the integration of primary public repositories to ensure chemical and bioactivity representativeness, aligning with foundational QSAR guidelines that mandate rigorous data preparation [33]:

- **Primary Data Sources:** The initial collection was constructed by combining records from BindingDB, which provided detailed drug-protein interaction parameters and SMILES structures, alongside bioactivity data retrieved from ChEMBL [32].

- **Consolidation and Curation Protocol:** Following database integration, discordant duplicates and intermediate compounds were systematically removed to mitigate experimental noise and avoid overfitting, adhering to internationally recognized QSAR curation standards [34, 35]. Automated standardization steps, such as salt stripping and structural validation, were applied to guarantee overall chemical integrity [15].

- **Treatment of Experimental Variations:** To address compounds with multiple IC_{50} measurements obtained under varying conditions, the Interquartile Range (IQR) method was employed to filter out statistical outliers across identical CID identifiers [32]. For the remaining consistent entries, an average IC_{50} value was calculated to establish a single representative measurement per ligand group [32].

- **Target Variable Formulation:** The final dataset comprises a benchmark of 873 unique molecules [32]. The continuous target variable was formulated as pIC_{50}, defined as the negative logarithm of the molar IC_{50} concentration ($pIC_{50} = -\log_{10}(IC_{50})$)

[32]. This logarithmic conversion standardizes the response values, reduces variance, and compresses the active range, rendering it optimal for robust regression algorithms [32].

2.2 Representation Spaces: Molecular Fingerprints and Conformational Descriptors

To systematically evaluate the structure-activity relationship, three distinct molecular representation spaces were programmatically generated using the open-source cheminformatics library RDKit [36]:

- **Molecular Fingerprints (ECFP4):** Morgan fingerprints were computed to represent molecular structures as vector spaces [37]. Extended-connectivity fingerprints (ECFPs) were specifically developed for structure-activity modeling because they can be calculated rapidly and represent an essentially infinite number of distinct molecular features, including stereochemical information [38]. Furthermore, classical 2D fingerprint-based models have been demonstrated to perform as well as state-of-the-art 3D structure-based models for bioactivity predictions relying solely on ligand information [39].

- **2D Physicochemical Descriptors:** A comprehensive set of conventional two-dimensional topological representations was extracted to establish a quantitative physicochemical baseline [37]. Well-curated 2D molecular representations are highly robust, and recent studies have shown they can maintain scaffold-robust advantages over 3D alternatives in various permeability and property prediction tasks [40].

- **3D Conformational Descriptors (ETKDG):** Three-dimensional conformers were generated employing the distance geometry approach implemented in RDKit [41]. Because purely distance-based constraints tend to lead to distorted aromatic rings and sp^2 centers, this methodology utilizes experimental torsion-angle preferences obtained from small-molecule crystallographic data to improve conformation generation [41]. Although recent evaluations indicate that 3D conformer descriptors may provide limited incremental predictive value over curated 2D representations in specific tabular benchmark algorithms [40], integrating spatial information remains highly relevant for molecular image-based modeling [42] and geometry-augmented representation learning [43].

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